

1 Sampling and Sampling Distributions

We mentioned that statistics can be subdivided into areas of descriptive statistics, probability theory and statistical inference. In probability theory, we assume that we know how a specific population looks like, and we try to answer questions like: what is the probability that some specific event happens? Or, in other words, if we observe a number of elements from the population (a sample), what is the probability that this sample has some specific characteristics? For example, if the population consists of all males in Lesotho and we know that 10% of that population is larger than 1.78 metres, then we can answer the question: what is the probability that among 100 males selected at random, there will be at least 20 who are taller than 1.78m?

In statistical inference, we reverse this process. Now we assume that we do **not** know how the population looks like, but we try to make statements about the population based on a number of observations we make (the sample). We could, for example, observe that 20 out of a sample of 100 males are taller than 1.78m. What conclusion can we draw in that case about the proportion of males taller than 1.78m in the whole population?

Any conclusion made in such situation will involve some degree of uncertainty. However, if the observations were made following certain principles, then the degree of uncertainty can be measured. The theory of statistics can help us by providing techniques for making inductive inference and for measuring the degree of uncertainty of such inferences. Because uncertainty is measured in terms of probability, we will draw on the probability theory as presented earlier.

1.1 Random samples and statistics

A **population** is the totality of elements which are under study and about which information is desired. There are two ways we can collect data to study the population characteristics of interest: inspect all population units or inspect a portion of them. When we inspect each and every unit in the population, we say that a **census** has been taken. Unless the population size is small, this rarely occurs, due to costs and time considerations. So in most cases, we examine only a part of the whole population.

A **sample** is a subset of a population chosen to draw inferences (conclusions) about characteristics of that population. The quality of inferences drawn from the sample data is directly related to how well the sample represent the population. There is virtually no way, apart from investigating the entire population (taking a census), that we can guarantee that the sample will perfectly represent the population. So it is important to pay careful attention to the way a sample is selected.

We assume that each element in our population has some numerical value associated with it, representing the characteristic about which we would like to obtain information. For example, the population may be student population at NUL and the numerical value the age of a student; or the population may be males in Lesotho and the numerical value is 1 if he is taller than 1.78m, and 0

otherwise. These numerical values are, of course, random variables with certain probability mass or density functions.

Suppose we take a sample of size n , $n > 1$, and let the random variable X_i represent the numerical value that the i th sample item will assume. If

1. all X_i have the same probability function $f(\cdot)$ and
2. all X_i are independent from each other, that is $f_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n) = f(x_1)f(x_2) \dots f(x_n)$,

then $X_1, X_2, \dots, X_n$ is called a **random sample** of size n drawn from a population with probability function $f(\cdot)$. We will often shorten this by saying that $X_1, X_2, \dots, X_n$ is a random sample from probability function $f(\cdot)$.

This definition, strictly speaking, only applies to situations where the population is infinitely large or when sampling is done from finite populations with replacement. In the case of sampling from a finite population without replacement, it can be shown that the random variables X_i are never independent. Therefore, another definition is used for random sampling from a finite population, that is that each possible set of a specified size must have an equal probability of being selected. In the remainder of the course, we will assume that the population under investigation is infinite in size. Even when this is not strictly true for many applications, the population size is then assumed to be large enough to justify the treatment of the population as an infinite population.

Definition 1.1 A *statistic* is a function of the sample random variables $X_1, X_2, \dots, X_n$ which contains no unknown values.

For example, $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is a statistic, just as $S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$. However, $\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$ is not a statistic unless the value of μ is known already. Thus, if we take a random sample of size n and observe the resulting values $x_1, x_2, \dots, x_n$ we can always compute the value of a statistic. It is important to stress that a statistic is itself a random variable with a certain probability distribution. Suppose we take many times a random sample of size n and each time we compute the value of the statistic. Then it become apparent that some values of the statistic are more likely to occur than others, in other words the statistic has a specific probability distribution. In some cases, we will be able to determine the exact probability distribution of the statistic; in other cases, we can only give an approximate distribution. The probability distribution of a statistic is called its **sampling distribution**.

1.1.1 Sample mean

Let $X_1, X_2, \dots, X_n$ be a random sample from a probability distribution with mean μ and variance σ^2 . The **sample mean**, denoted by $\bar{X}$, is defined by:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad (1)$$

After the sample is actually drawn and the values $x_1, x_2, \dots, x_n$ are known numerically, the sample mean is the value:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \dots + x_n}{n}. \quad (2)$$

The random variable $\bar{X}$ has mean

$$E(\bar{X}) = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E(X_i) = \mu \quad (3)$$

and variance

$$\sigma_{\bar{X}}^2 = \text{Var}(\bar{X}) = \text{Var}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{\sigma^2}{n}. \quad (4)$$

Therefore, the standard deviation of $\bar{X}$ is $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$. This shows that $\bar{X}$ has the distribution with mean equal to the population mean μ , and that the spread of the values of $\bar{X}$ about μ is small for a large sample size as compared to a small sample size. This means that in theory we can make the variance of the sample mean as small as desired.

1.1.2 Sample variance

Let $X_1, X_2, \dots, X_n$ be a random sample of size n . The sample variance, denoted by S^2 , is defined as:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2. \quad (5)$$

Check that S^2 can be written as

$$S^2 = \frac{1}{n-1} \left[\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right],$$

which is easier to use for computations. Thus, the sample standard deviation is just

$$S = \sqrt{\frac{1}{n-1} \left[\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right]}.$$

The expected value of the random variable S^2 is

$$\begin{aligned}
E(S^2) &= E\left\{\frac{1}{n-1}\left[\sum_{i=1}^n X_i^2 - n\bar{X}^2\right]\right\} \\
&= \frac{1}{n-1}\left[\sum_{i=1}^n E(X_i^2) - nE(\bar{X}^2)\right] \\
&= \frac{1}{n-1}\left[\sum_{i=1}^n (\mu^2 + \sigma^2) - n\left(\mu^2 + \frac{\sigma^2}{n}\right)\right] \\
&= \frac{1}{n-1}[n\mu^2 + n\sigma^2 - n\mu^2 - \sigma^2] \\
&= \frac{1}{n-1}(n\sigma^2 - \sigma^2) = \frac{\sigma^2}{n-1}(n-1) = \sigma^2
\end{aligned}$$

This shows that the sampling distribution of S^2 is centred on σ^2 , the population variance. We call a statistic whose expected value equals the parameter being estimated an **unbiased** estimator (statistic). Thus, $\bar{X}$ and S^2 as defined above are unbiased statistics for parameters μ and σ^2 , respectively.

1.2 Sampling from a normal distribution

Up to now, we have not said anything about the form of the population being sampled, only that it has mean μ and variance σ^2 . As a result, the probability distribution of $\bar{X}$ and S^2 could not be determined. However, if sampling is made from the normal distribution, then exact distribution of these random variables can be found. We first consider the sampling distribution of $\bar{X}$.

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a normal population with mean μ and variance σ^2 . This implies that $X_i \sim N(\mu, \sigma^2)$, for $i = 1, 2, \dots, n$. Now define the sample mean as

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

One important property of normal random variables is that any linear function of normal random variables is itself a normal random variable. We see that $\bar{X}$ is simply a linear function of n normal random variables. As such, $\bar{X}$ must be a normal random variable. That is,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad (6)$$

and thus

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1). \quad (7)$$

We can, therefore evaluate probability statements about the random variable $\bar{X}$ by transforming to standard normal random variable using Equation (7).

Example 1.1 Suppose we take a random sample of size $n = 25$ from a normal population with mean $\mu = 3$ and $\sigma = 0.5$. What is the probability that the sample mean of these 25 observations will be

- a) larger than 3.15;
- b) less than 2.9;
- c) between 2.8 and 2.95?

Solution Since we know that the sample mean is normally distributed with mean $\mu_{\bar{X}} = 3$ and standard deviation $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{0.5}{\sqrt{25}} = 0.1$, we can standardize $\bar{X}$ by subtracting its mean and dividing by its standard deviation as follows:

a)

$$\begin{aligned} P(\bar{X} > 3.15) &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{3.15 - 3}{0.1}\right) \\ &= P(Z > 1.5) = 1 - P(Z \leq 1.5) \\ &= 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668 \end{aligned}$$

b)

$$\begin{aligned} P(\bar{X} < 2.9) &= P\left(Z < \frac{2.9 - 3}{0.1}\right) \\ &= P(Z < -1) = P(Z > 1) \\ &= 1 - \Phi(1) = 1 - 0.8413 = 0.1587 \end{aligned}$$

c)

$$\begin{aligned} P(2.8 < \bar{X} < 2.95) &= P\left(\frac{2.8 - 3}{0.1} < Z < \frac{2.95 - 3}{0.1}\right) \\ &= P(-2 < Z < -0.5) = P(0.5 < Z < 2) \\ &= \Phi(2) - \Phi(0.5) = 0.9773 - 0.6915 = 0.2858. \end{aligned}$$

Example 1.2 Suppose that X follows a normal distribution with mean μ and variance 1. Find the probability that X will lie within 0.3 units of the population mean μ .

Solution We are given that $X \sim (\mu, 1)$ and are asked to find $P(|X - \mu| \leq 0.3)$. We standardize as usual to get

$$\begin{aligned} P(|X - \mu| \leq 0.3) &= P(-0.3 \leq X - \mu \leq 0.3) \\ &= P\left(-\frac{0.3}{1} \leq \frac{X - \mu}{\sigma} \leq \frac{0.3}{1}\right) \\ &= P(-0.3 \leq Z \leq 0.3) = 2\Phi(0.3) - 1 \\ &= 2 \times 0.6179 - 1 = 0.2358. \end{aligned}$$

Let us continue with Example 1.2. Suppose now that we take a random sample of size $n = 9$ from this distribution. What is the probability that the sample mean of these 9 observations will be within 0.3 units of the population mean?

Solution Here the required probability is about $\bar{X}$ not about X . We know that $\bar{X} \sim N(\mu, \sigma^2/n)$ and we wish to find $P(|\bar{X} - \mu| \leq 0.3)$. This is

$$\begin{aligned} P(|\bar{X} - \mu| \leq 0.3) &= P(-0.3 \leq \bar{X} - \mu \leq 0.3) \\ &= P\left(-\frac{0.3}{1/\sqrt{9}} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \frac{0.3}{1/\sqrt{9}}\right) \\ &= P(-0.9 \leq Z \leq 0.9) \\ &= 2\Phi(0.9) - 1 = 2 \times 0.8159 - 1 = 0.6318. \end{aligned}$$

Exercise

- Compute the preceding probability for the following values of n : $n = 25$, $n = 36$. Compare the probabilities and comment.
- Now, suppose $n = 25$, compute the following probabilities: i) $P(|\bar{X} - \mu| \leq 0.2)$, ii) $P(|\bar{X} - \mu| \leq 0.4)$. Compare the probabilities: what do you realise?

Example 1.3 Next, suppose we would like the sample mean to lie within 0.1 units of the population mean with probability 0.95. What sample size n will ensure that this happens? That is, what is the value of n such that $P(\bar{X} - \mu| \leq 0.1) = 0.95$?

Solution To answer this question, we can proceed as follows:

$$\begin{aligned} P(|\bar{X} - \mu| \leq 0.1) &= 0.95 \\ P(-0.1 \leq \bar{X} - \mu \leq 0.1) &= 0.95 \\ P\left(-\frac{0.1}{1/\sqrt{n}} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \frac{0.1}{1/\sqrt{n}}\right) &= 0.95 \\ P(-0.1\sqrt{n} \leq Z \leq 0.1\sqrt{n}) &= 0.95 \\ 2\Phi(0.1\sqrt{n}) - 1 &= 0.95 \\ 2\Phi(z) - 1 &= 0.95, \end{aligned}$$

where $z = 0.1\sqrt{n}$. We have come across problems like this before, in which we are given area under the standard normal curve and we are to determine the corresponding z value. Following that procedure we see that $z = 1.96$. Therefore,

$$\begin{aligned} 0.1\sqrt{n} &= 1.96 \\ \sqrt{n} &= \frac{1.96}{0.1} \\ n &= \left(\frac{1.96}{0.1}\right)^2 = 384.16 \end{aligned}$$

This tells us that the value of n that will give us the required accuracy is 384.16, but since it is not possible to have a sample of size 384.16, we round this number up to the nearest integer. Therefore, $n = 385$ will ensure we get the required accuracy.

In general, suppose that X follows a $N(\mu, \sigma^2)$ population. What is the value of n such that the sample mean will lie within ε , $\varepsilon > 0$, units of the population mean with probability γ , $0 < \gamma < 1$? Following the procedure used in Example 1.3 shows that

$$n = \left(\frac{z_{(1+\gamma)/2} \sigma}{\varepsilon} \right)^2.$$

Example 1.4 Consider a normal population with unknown mean μ and variance 16. How large a sample should be taken to ensure that the probability is 0.9 that the sample mean will lie within 0.5 units of the population mean?

Solution Here we are given that $\gamma = 0.9$, $\sigma = 4$ and $\varepsilon = 0.5$. From the standard normal table we get $z_{(1+0.9)/2} = z_{0.95} = 1.645$. Substituting all this information in the formula, we get

$$n = \left(\frac{1.645 \times 4}{0.5} \right)^2 = 13.16^2 = 173.1855.$$

Rounding up to the nearest integer shows that the required sample size is $n = 174$.

1.3 The central limit theorem

In Section 1.2, we showed that if sampling is made from a normal population, then the sampling distribution of the sample mean $\bar{X}$ is normal as well. If sampling is made from a non-normal distribution, the sampling distribution of $\bar{X}$ is not normal. However, the following result, called the **central limit theorem**, gives us the approximate sampling distribution of $\bar{X}$ when the sample size is large enough.

Theorem 1.1 *Let the population have probability function $f(\cdot)$ with mean μ and variance σ^2 , and let $\bar{X}$ be the sample mean of a random sample of size n from $f(\cdot)$. Now define a random variable Z_n by*

$$Z_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

Then, Z_n has a distribution that approaches the standard normal distribution as $n \rightarrow \infty$.

We will not attempt the proof of this theorem here. However, the theorem tells us that the sampling distribution of $\bar{X}$ will become increasingly close to a normal distribution as the sample size increases. The remarkable thing is that nothing is said about the form of the population distribution, $f(\cdot)$, only that the mean μ and variance σ^2 exist. The importance of this theorem derives from

the fact that it enables us to make and evaluate probability statements about the distribution of the sample mean, even when the population distribution is not normal. How large the sample size should be depends on the form of the population distribution $f(\cdot)$. If $f(\cdot)$ is itself normal, $\bar{X}$ is normally distributed for any value of the sample size.

Example 1.5 The number of persons entering a bank during 5-minute intervals is known to have a Poisson distribution with mean $\lambda = 2$ (For a Poisson random variable the mean equals the variance). If 100 of these 5-minute intervals are randomly selected and the number of arrival in each interval is recorded, what is the probability the sample mean number of arrivals will be less than 1.85?

Solution The sample size of 100 is large enough to apply the central limit theorem. Thus, the sample mean $\bar{X}$ will be approximately normally distributed with mean 2 and standard deviation $\sqrt{2/100} = 0.1414$. Therefore,

$$\begin{aligned} P(\bar{X} \leq 1.85) &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \frac{1.85 - 2}{0.1414}\right) \\ &\approx P(Z \leq -1.06) = P(Z > 1.06) \\ &= 1 - \Phi(1.06) = 1 - 0.8554 = 0.1446. \end{aligned}$$

Example 1.6 The service times for customers coming through a checkout counter in a retail store are independent random variables with mean 1.5 minutes and variance 1. Approximate the probability that 100 customers can be serviced in less than 2 hours of the total service time.

Solution Let $X_i, i = 1, 2, \dots, 100$, denote the service time for the i th customer, then we require the probability

$$P\left(\sum_{i=1}^{100} X_i \leq 120\right) = P\left(\frac{\sum_{i=1}^{100} X_i}{100} \leq \frac{120}{100}\right) = P(\bar{X} \leq 1.2).$$

Because the sample size is large, the central limit theorem tells us that $\bar{X}$ is approximately $N(1.5, 1/100)$. Therefore,

$$\begin{aligned} P(\bar{X} \leq 1.2) &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \frac{1.2 - 1.5}{1/\sqrt{100}}\right) \\ &= P(Z \leq -3) = 1 - \Phi(3) \\ &= 1 - 0.9987 = 0.0013. \end{aligned}$$

This shows that it is highly unlikely that 100 customers can be serviced in less than 2 hours of total service time.

1.4 Sampling distribution of S^2

Suppose that $X_1, X_2, \dots, X_n$ is a random sample from a normal population with mean μ and variance σ^2 . We have defined the sample variance as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Now, let us write

$$(n-1)S^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Dividing by σ^2 throughout gives

$$\begin{aligned} \frac{(n-1)S^2}{\sigma^2} &= \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} - \frac{n(\bar{X} - \mu)^2}{\sigma^2} \\ &= \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 - \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2. \end{aligned}$$

We notice that the two random variable on the right hand side of the last equality are chi-square random variables with respective degrees of freedom n and 1. Since $\bar{X}$ and S^2 are independent, we can deduce that

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2. \quad (8)$$

This makes it possible to make and evaluate probability statements about S^2 (or S). For example, we evaluate statements like

1. $P(a < S < b)$,
2. $P(S < b)$,
3. $P(S > b)$, etc,

where a and b are positive numbers such that $a < b$. To do this, we write

$$\begin{aligned} P(S < b) &= P(S^2 < b^2) = P\left(\frac{(n-1)S^2}{\sigma^2} < \frac{(n-1)b^2}{\sigma^2}\right) \\ &= P(\chi_{n-1}^2 < d), \end{aligned}$$

where $d = \frac{(n-1)b^2}{\sigma^2}$. We then look up the probability in the chi-square table with $n-1$ degrees of freedom.

Example 1.7 Suppose that $X_1, X_2, \dots, X_9$ is a random sample from a normal population with mean μ and variance $\sigma^2 = 25$. What is the probability that the standard deviation of these 9 observations will be larger than 7?

Solution We are asked to find $P(S > 7)$. Proceeding as shown above, we have

$$\begin{aligned} P(S > 7) &= P(S^2 > 49) = P\left(\frac{(n-1)S^2}{\sigma^2} > \frac{8 \star 49}{25}\right) \\ &= P(\chi_8^2 > 15.68) \approx P(\chi_8^2 > 15.507) \\ &= 0.05. \end{aligned}$$

Or, we can give bounds

$$0.025 < P(S > 7) < 0.05.$$

1.5 Sampling Distribution of Sample Proportion

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a Bernoulli population (distribution) with parameter p , the probability of success. Then,

$$X_i = \begin{cases} 0, & \text{if failure} \\ 1, & \text{if success.} \end{cases}$$

Then,

$$X = \sum_{i=1}^n X_i$$

is a binomial random variable with parameters n and p . The sample mean of the observations, $X_1, X_2, \dots, X_n$, is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{X}{n},$$

which is the sample proportion. This can be used to estimate the population proportion (probability of success), p . The expected value (mean) of $\bar{X}$ is

$$E(\bar{X}) = E\left[\frac{X}{n}\right] = \frac{1}{n}E(X) = p,$$

which is the parameter being estimated. The variance of $\bar{X}$ is

$$\text{Var}(\bar{X}) = \text{Var}\left[\frac{X}{n}\right] = \frac{p(1-p)}{n}$$

and the standard deviation or *standard error* of $\bar{X}$, $SE(\bar{X})$ is $SE(\bar{X}) = \sqrt{\frac{p(1-p)}{n}}$.

Since $\bar{X}$ is simply the value of X expressed as a proportion, the sampling distribution of $\bar{X}$ is identical to the probability distribution of X , except that it has a new scale along the horizontal axis (ie, the range). Recall that the range for X is

$$R_X = \{0, 1, 2, \dots, n\}.$$

This means that range for $\bar{X}$ is

$$R_{\bar{X}} = \{0, \frac{1}{n}, \frac{2}{n}, \dots, 1\}.$$

For instance, let $n = 5$ and $p = 0.5$, then $R_X = \{0, 1, 2, 3, 4, 5\}$ and $R_{\bar{X}} = \{0, \frac{1}{5}, \frac{2}{5}, \dots, 1\}$. Then, $P(X = 0) = P(\bar{X} = 0)$, $P(X = 1) = P(\bar{X} = \frac{1}{5})$, $P(X = 2) = P(\bar{X} = \frac{2}{5})$, etc.

When the sample size n is large, then by the **Central Limit Theorem**,

$$\bar{X} \sim N\left(p, \frac{p(1-p)}{n}\right)$$

and

$$Z = \frac{\bar{X} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0, 1).$$

So all probability statements about $\bar{X}$ can be evaluated by first standardizing and referring to the standard normal table as usual.

Example 1.8 Candidate A believes that she can win a city election if she can earn at least 55% of the votes in precinct 1. She also believes that about 50% of the city's voters favor her. If $n = 100$ voters show up to vote at precinct 1, what is the probability that candidate A will receive at least 55% of their votes?

Solution Let X denote the number of voters at precinct 1 who vote for candidate A. We must approximate $P(\bar{X} = X/n \geq 0.55)$ when p , the probability that a randomly selected voter favors candidate A is 0.5. If we think of the $n = 100$ voters at precinct 1 as a random sample from the city, then X has a binomial distribution with $n = 100$ and $p = 0.5$. Because $\bar{X}$ satisfies the Central Limit Theorem, we know that $\bar{X}$ is approximately normally distributed with mean $p = 0.5$ and variance $\frac{p(1-p)}{n} = \frac{(0.5)(0.5)}{100} = 0.0025$. Therefore,

$$P(\bar{X} \geq 0.55) = P\left(\frac{\bar{X} - p}{\sqrt{\frac{p(1-p)}{n}}} \geq \frac{0.55 - 0.5}{0.05}\right) \approx P(Z \geq 1) = 0.1587.$$

Exercise An airline finds that 5% of the persons who make reservations on a certain flight do not show up for the flight. If the airline sells 160 tickets for a flight with only 155 seats, what is the probability that a seat will be available for every person holding a reservation and planning to fly?

2 Estimation

Suppose that we desire to study a population which has probability distribution $f(\cdot : \theta)$, where the form of the distribution is known, but it contains an unknown parameter θ . (Note that θ could be a vector, for example, if the population is normal with both mean μ and variance σ^2 unknown, then $\theta = (\mu, \sigma^2)$).

The procedure is to take a random sample of size n , $X_1, X_2, \dots, X_n$, from this distribution and let some statistic represent or estimate this unknown parameter.

Definition 2.1 An **estimator** is a rule, often expressed as a formula, that tells us how to calculate an **estimate** based on information in the sample. By definition, an estimator is a statistic.

Point Estimation Based on the sample data, a single number is calculated to estimate the population parameter. The rule or formula that describes this calculation is called the point estimator and the resulting number is called the point estimate.

Interval Estimation Based on the sample data, two numbers are calculated to form an interval within which the parameter is expected to lie. The rule or formula that describes this calculation is called interval estimator, and the resulting numbers are called interval estimate or confidence interval.

Both point and interval estimation procedures use information provided by the sampling distribution of the specific estimator you have chosen.

2.1 Point Estimation

Sampling distributions provide information that can be used to select the best estimator. What characteristics would be valuable for an estimator? First, the sampling distribution of the point estimator should be centred over the true value of the parameter to be estimated.

Definition 2.2 *An estimator is said to be **unbiased** if the mean of its sampling distribution is equal to the true value of the parameter being estimated.*

That is, if $\hat{\theta}$ is an estimator for the parameter θ , then $\hat{\theta}$ is unbiased if and only if $E(\hat{\theta}) = \theta$. The second desirable characteristic of an estimator is that the spread (variance) of the sampling distribution should be as small as possible. This ensures that, with high probability, an individual estimate will fall close to the true value of the parameter.

2.1.1 Error of estimation

The distance between the estimator and the true value of the parameter is called the **error of estimation**. We will denote it by ϵ . That is, if $\hat{\theta}$ is an estimator for parameter θ , then

$$\epsilon = |\hat{\theta} - \theta|.$$

Because $\hat{\theta}$ is a random variable, the error of estimation is also a random variable and we cannot say how small or large it will be for a particular estimate. However, we can make probability statements about it.

Note that, when estimating the mean μ of the normal population, then $\epsilon = |\bar{X} - \mu|$ and for estimating population proportion $\epsilon = |\bar{X} - p|$, etc.

When the sample size is large, then the sampling distribution of $\bar{X}$ (for estimating μ and population proportion, p) is approximately normal by the Central Limit Theorem. The empirical rule states that 95% of all point estimates will lie within 1.96 standard deviations of the mean of the distribution. For unbiased estimators, this implies that the difference between the point estimator and the true value of the parameter will be less than 1.96 Standard deviations or 1.96 standard errors. This quantity is called the 95% **margin of error** and it provides a practical upper bound for the error of estimation. It is possible that the error of estimation will exceed this margin of error, but this is very unlikely.

For estimating the population mean μ for a quantitative population, the point estimator $\bar{X}$ is unbiased with standard error

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The $100\gamma\%$ margin of error when n is large is

$$MOE = z_{(1+\gamma)/2} \frac{\sigma}{\sqrt{n}}.$$

When σ is not known, it is estimated from the data by $S = \sqrt{S^2}$ and the margin of error becomes

$$MOE = z_{(1+\gamma)/2} \frac{S}{\sqrt{n}}.$$

To estimate the population proportion, p , for a binomial population, the point estimator $\bar{X} = \frac{X}{n}$ is unbiased with standard error

$$SE(\bar{X}) = \sqrt{\frac{p(1-p)}{n}}.$$

Since p is not known, we estimate it by $\bar{X}$ and the estimated standard error becomes

$$SE(\bar{X}) = \sqrt{\frac{\bar{X}(1-\bar{X})}{n}}.$$

The $100\gamma\%$ margin of error is estimated as

$$MOE = z_{(1+\gamma)/2} \sqrt{\frac{\bar{X}(1-\bar{X})}{n}}.$$

Example 2.1 Suppose that a random sample of size $n = 50$ gave the mean of 11.5 units and standard deviation of 3.5 units. Use this information to estimate the population mean μ and place a 98% margin of error on this estimate.

Solution The point estimate for μ is $\bar{x} = 11.5$. The 98% margin of error is estimated as

$$MOE = z_{(1+0.98)/2} \frac{S}{\sqrt{n}}$$

where $z_{(1+0.98)/2} = z_{0.99} = 2.326$, $s = 3.5$ and $n = 50$. Substituting we get

$$MOE = 2.326 \star \frac{3.5}{\sqrt{50}} = 1.15.$$

Example 2.2 A random sample of $n = 900$ observations from a binomial population produced $x = 665$ successes. Estimate the binomial proportion (probability of success) p and calculate:

- a) the 95% margin of error;

- b) the 98% margin of error;
- c) the 99% margin of error.

Comment on the differences between margin of errors.

Solution Exercise!!

3 Interval Estimation

An interval estimator is a rule specifying the method for using sample measurements to calculate two numbers that form the endpoints of the interval. Ideally, the resulting interval will have two properties: first, it will contain the target parameter; second, it will be relatively narrow. One or both of the endpoints of the interval, being functions of sample measurements, will vary randomly from sample to sample. Thus, the length and location of the interval are random quantities, and we cannot be certain that the target parameter θ will fall between the endpoints of any single interval calculated from a single sample. This being the case, our objective is to find an interval estimator capable of generating narrow intervals that have high probability of enclosing θ .

Interval estimates are commonly called *confidence intervals*. The upper and lower endpoints of the confidence interval are called upper and lower confidence limits. The probability that the confidence interval will enclose θ is called a confidence coefficient, denoted by γ , where $0 < \gamma < 1$. From practical point of view, the confidence coefficient identifies the fraction of the time, in repeated sampling, that the interval constructed will contain the target parameter θ .

Suppose that $\hat{\theta}_L$ and $\hat{\theta}_U$ are the lower and upper confidence limits, respectively, for parameter θ . Then, if

$$P(\hat{\theta}_L \leq \theta \leq \hat{\theta}_U) = \gamma, \quad (9)$$

the probability γ is the confidence coefficient. The resulting confidence interval defined by $[\hat{\theta}_L, \hat{\theta}_U]$ is called a two sided confidence interval.

It is also possible to form a one-sided confidence interval such that

$$P(\hat{\theta}_L \leq \theta) = \gamma. \quad (10)$$

The resulting confidence interval is $[\hat{\theta}_L, \infty)$. This is called a lower-limit confidence interval. Similarly, we could form an upper one-sided confidence interval such that

$$P(\theta \leq \hat{\theta}_U) = \gamma. \quad (11)$$

The implied confidence interval is $(-\infty, \hat{\theta}_U]$.

3.1 Methods for Finding Confidence Intervals

A very useful method for finding confidence intervals is called the *pivotal quantity method*. This method depends upon finding a pivotal quantity that possesses two characteristics:

1. It is the function of the sample measurements and the unknown parameter θ , where θ is the only unknown quantity.
2. Its probability distribution does not depend on the unknown parameter θ .

If the probability distribution of the pivotal quantity is known, then we can use the following logic: if Y is a random variable, c a constant ($c > 0$), and $P(a \leq Y \leq b) = \gamma$, then $P(ca \leq cY \leq cb) = \gamma$. Similarly, for any constant d , $P(a + d \leq Y + d \leq b + d) = \gamma$, that is, the probability of the event $a \leq Y \leq b$ is unaffected by change of scale or translation of Y .

We illustrate this method using the following example.

Example 3.1 Suppose that a random sample of size $n = 4$ is taken from a normal population with unknown mean μ and standard deviation $\sigma = 3$. Find a 95% confidence interval for μ .

Solution We have seen that the unbiased estimator for μ with relatively small variance is $\bar{X}$. Here, the confidence coefficient is $\gamma = 0.95$. Furthermore, we know that

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Since Z has the two characteristics mentioned above, it can be used as a pivotal quantity. We also know that

$$P(-1.96 \leq Z \leq 1.96) = 0.95.$$

Now, substituting for Z in the middle of the inequality, we get

$$P\left(-1.96 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq 1.96\right) = 0.95.$$

By some algebra, this probability statement can be written as

$$P\left(\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}\right) = 0.95.$$

Notice that this probability statement is of the form of equation (1) above, so the 95% confidence interval is

$$\left[\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}\right].$$

Suppose now that the observed sample is: $x_1 = 1.2, x_2 = 3.4, x_3 = 0.6, x_4 = 5.6$. We can then compute the observed sample mean as $\bar{x} = 2.7$. Substituting this together with the other information given in the above, we get the observed confidence interval

$$\left[2.7 - 1.96 \star \frac{3}{\sqrt{4}}, 2.7 + 1.96 \star \frac{3}{\sqrt{4}}\right] = [-0.24, 5.64].$$

The interval width is computed by taking the difference between the upper and lower limits, so in this case the width is $5.64 - (-0.24) = 5.88$.

Similarly, confidence interval with any degree of confidence can be obtained. Thus, since

$$P(-2.326 \leq Z \leq 2.326) = 0.99,$$

a 99% confidence interval is $[-0.789, 6.189]$. Notice, however, that this confidence interval is wider than the previous 95% interval, which of course is as should be.

It is possible to construct many different confidence intervals with the same confidence coefficient. For example, since $P(-1.68 \leq Z \leq 2.70) = 0.95$ and $P(-1.66 \leq Z \leq 2.96) = 0.95$, two other 95% confidence intervals are $(-1.35, 5.22)$ and $(-1.74, 5.19)$, respectively. However, both these intervals are inferior to the first one because their width are larger, meaning they give less precise information about the location of μ . Any two numbers a and b such that $P(a \leq Z \leq b) = 0.95$ will determine a 95% confidence interval. We would like the confidence interval to be as short as possible. Since the standard normal density is symmetrical about $z = 0$, it can be shown that this occurs when we take $a = -b$.

We can find confidence interval for any value of the confidence coefficient γ between 0 and 1, that is, $0 < \gamma < 1$. The resulting confidence interval is then a $100\gamma\%$ confidence interval.

The confidence interval that we will now discuss apply strictly speaking only when sampling from a normal population. However, the Central Limit Theorem tells us that when sample size is large enough (say $n > 30$), the sample mean is approximately normally distributed. So large-sample confidence intervals can be developed when a sampled population is not normal.

3.1.1 Confidence interval for the mean when σ^2 is known

This was the case of the example previously considered. Suppose $X_1, X_2, \dots, X_n$ is a random sample from a normal population with unknown mean μ and known variance σ^2 . To find a $100\gamma\%$ confidence interval for μ , we use

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

as a pivotal quantity. We then use the standard normal table to find the value of z such that $P(-z \leq Z \leq z) = \gamma$. Since

$$P(-z \leq Z \leq z) = \Phi(z) - \Phi(-z) = 2\Phi(z) - 1$$

it follows that we must find z such that $\Phi(z) = (1 + \gamma)/2$. Therefore, z is the $(1 + \gamma)/2$ -th percentile point of the standard normal distribution, denoted as $z_{(1+\gamma)/2}$. After determining $z_{(1+\gamma)/2}$ we can substitute Z by $(\bar{X} - \mu)/\sigma/\sqrt{n}$ to get

$$P\left(-z_{(1+\gamma)/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{(1+\gamma)/2}\right) = \gamma$$

which can be written as

$$P\left(\bar{X} - z_{(1+\gamma)/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{(1+\gamma)/2} \frac{\sigma}{\sqrt{n}}\right) = \gamma.$$

Thus,

$$\left[\bar{X} + z_{(1-\gamma)/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{(1+\gamma)/2} \frac{\sigma}{\sqrt{n}}\right]$$

is a $100\gamma\%$ confidence interval for μ . After observing the values in a random sample, we can substitute $\bar{X}$ by the observe sample mean $\bar{x}$ to get

$$\left[\bar{x} + z_{(1-\gamma)/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{(1+\gamma)/2} \frac{\sigma}{\sqrt{n}}\right]$$

Note that it would not be correct to say that

$$P\left[\bar{x} + z_{(1-\gamma)/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{(1+\gamma)/2} \frac{\sigma}{\sqrt{n}}\right] = \gamma$$

Example 3.2 Suppose that a random sample of size $n = 40$ from a normal population with unknown mean μ and variance σ^2 , resulted in $\bar{x} = 7.164$. find an 80% confidence interval for μ .

Solution We know that a $100\gamma\%$ confidence interval for μ is given by

$$\left[\bar{x} + z_{(1-\gamma)/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{(1+\gamma)/2} \frac{\sigma}{\sqrt{n}}\right].$$

We are given that $\bar{x} = 7.164$, $\sigma = \sqrt{10}$, $n = 40$ and from the standard normal table we get $z_{(1+0.8)/2} = z_{0.9} = 1.282$. Substituting, we get an 80% interval estimate for μ as

$$\left[7.164 - 1.282 \star \frac{3.163}{\sqrt{40}}, 7.164 + 1.282 \star \frac{3.163}{\sqrt{40}}\right] = [6.523, 7.805].$$

4 Large Sample Confidence Intervals

Suppose that $\hat{\theta}$ is an unbiased estimator for parameter θ , that is, $E(\hat{\theta}) = \theta$. Furthermore, let $\sigma_{\hat{\theta}}$ be the standard error of $\hat{\theta}$. Then, for large sample sizes,

$$Z = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}}$$

possesses approximately a standard normal distribution. Consequently, $Z = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}}$ forms (at least approximately) a pivotal quantity, and the pivotal method can be employed to develop confidence intervals for the target parameter θ . So to construct a $100(1 - \alpha)\%$ confidence interval for θ we use

$$Z = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}}$$

as the pivotal quantity. We start with the probability statement

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha.$$

Substituting in the middle of the inequality gives

$$P(-z_{\alpha/2} \leq \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \leq z_{\alpha/2}) = 1 - \alpha.$$

This can be written as

$$P(\hat{\theta} - z_{\alpha/2}\sigma_{\hat{\theta}} \leq \theta \leq \hat{\theta} + z_{\alpha/2}\sigma_{\hat{\theta}}) = 1 - \alpha,$$

and the implied confidence interval is

$$(\hat{\theta} - z_{\alpha/2}\sigma_{\hat{\theta}}, \hat{\theta} + z_{\alpha/2}\sigma_{\hat{\theta}}).$$

By analogy, we can determine 100(1-)% one-sided confidence limits, lower and upper bounds, respectively, to be

$$100(1 - \alpha)\% \text{ lower bound } \hat{\theta} - z_{\alpha}\sigma_{\hat{\theta}}$$

$$100(1 - \alpha)\% \text{ upper bound } \hat{\theta} + z_{\alpha}\sigma_{\hat{\theta}}$$

For estimating the parameters μ , p , $\mu_1 - \mu_2$ and $p_1 - p_2$, the following table shows the estimators and their corresponding standard errors:

Target parameter	Sample size(s)	Point estimator		Standard error
θ		$\hat{\theta}$	$E(\hat{\theta})$	$\sigma_{\hat{\theta}}$
μ	n	$\bar{X}$	μ	$\frac{\sigma}{\sqrt{n}}$
p	n	$\bar{X}$	p	$\sqrt{\frac{p(1-p)}{n}}$
$\mu_1 - \mu_2$	n_1 and n_2	$\bar{X}_1 - \bar{X}_2$	$\mu_1 - \mu_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
$p_1 - p_2$	n_1 and n_2	$\bar{X}_1 - \bar{X}_2$	$p_1 - p_2$	$\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$

The following examples illustrate applications of the general method developed here:

Example 4.1 The shopping times of $n = 64$ randomly selected customers at a local supermarket were recorded. The average and variance of the 64 shopping times were 33 minutes and 256 minutes², respectively. Estimate μ , the true average shopping time per customer, with a confidence coefficient of $1 - \alpha = 0.90$.

Solution In this case, we are interested in the parameter $\theta = \mu$. Thus, $\hat{\theta} = \bar{x} = 33$, and $s^2 = 256$ for a sample of $n = 64$ shopping times. The population variance σ^2 is unknown, so we use s^2 as its estimated value. The confidence interval

$$\hat{\theta} \pm z_{\alpha/2}\sigma_{\hat{\theta}}$$

has the form

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \approx \bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}.$$

From the standard normal table, we get $z_{\alpha/2} = z_{0.05} = 1.645$. Substituting into the formula, we get 90% confidence interval for μ as

$$33 \pm 1.645 \left(\frac{16}{8} \right) = (29.71, 36.29).$$

In repeated sampling, approximately 90% of all intervals of the form $\bar{X} \pm 1.645(S/\sqrt{n})$ include μ , the true mean shopping time per customer. Although we do not know whether the particular interval (29.71, 36.29) contains μ , the procedure that generated it yields intervals that do capture the true mean in approximately 90% of all instances where the procedure is used.

Example 4.2 Two independent random samples of size n_1 and n_2 were selected from populations 1 and 2. The sample means and sample variances were obtained as follows:

$$\begin{aligned}\bar{x}_1 &= 12.7, & \bar{x}_2 &= 7.4 \\ s_1^2 &= 1.38, & s_2^2 &= 4.14.\end{aligned}$$

Find a 95% confidence interval for the true difference in the two population means, $\mu_1 - \mu_2$. Based on the confidence interval, can you conclude that there is a difference in the two populations means?

Solution In this example, the confidence interval

$$\hat{\theta} \pm z_{\alpha/2} \sigma_{\hat{\theta}}$$

has the form

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}.$$

The population variances are not known but when sample sizes are large ($n_i > 30$), then the sample variances are good estimates of populations variances. So substituting the sample variances in the formula, we get

$$\begin{aligned}[(12.7 - 7.4) \pm 1.96 \sqrt{1.38/35 + 4.14/49}] &= [5.3 \pm 1.96 \sqrt{0.1239}] \\ &= [5.3 \pm 0.69] = [4.61, 5.99].\end{aligned}$$

Yes, the confidence interval suggests that $\mu_1 > \mu_2$.

Example 4.3 Two brands of refrigerators, denoted A and B, are each guaranteed for 1 year. In a random sample of 50 refrigerators of brand A, 12 were observed to fail before the guarantee period ended. An independent random sample of 60 brand B refrigerators also revealed 12 failures during the guarantee period. Estimate the true difference ($p_1 - p_2$) between proportions of failures during the guarantee period, with confidence coefficient approximately 0.98.

Solution The confidence interval

$$\hat{\theta} \pm z_{\alpha/2} \sigma_{\hat{\theta}}$$

now has the form

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{\bar{x}_1(1 - \bar{x}_1)}{n_1} + \frac{\bar{x}_2(1 - \bar{x}_2)}{n_2}}.$$

Because p_1 , and p_2 are unknown, the exact value of $\sigma_{\hat{\theta}}$ cannot be evaluated. But we can get a good approximation by substituting $\bar{x}_1$ for p_1 and $\bar{x}_2$ for p_2 , respectively.

For this example, $\bar{x}_1 = \frac{x_1}{n_1} = \frac{12}{50} = 0.24$, $\bar{x}_2 = \frac{x_2}{n_2} = \frac{12}{60} = 0.20$, and $z_{0.01} = 2.33$. Substituting, we get 98% confidence interval

$$(0.24 - 0.20) \pm 2.33 \sqrt{0.24 * 0.76 / 50 + 0.2 * 0.8 / 60} = 0.04 \pm 0.1852 = (-0.1452, 0.2252).$$

Notice that this confidence interval contains zero. Thus, a zero value for the difference in proportions ($p_1 - p_2$) is “believable” (at approximately the 98% confidence level) on the basis of the observed data. However, the interval also includes the value 0.1. Thus, 0.1 represents another value of ($p_1 - p_2$) that is “believable” on the basis of the data that we have analyzed.

5 Small Sample Confidence Intervals for μ and

$$\mu_1 - \mu_2$$

We assume that $X_1, X_2, \dots, X_n$ represent a random sample selected from a normal population, and we let $\bar{X}$ and S^2 represent the sample mean and sample variance, respectively. We would like to construct a confidence interval for the population mean when $\text{Var}(X_i) = \sigma^2$ is unknown and the sample size is too small to permit us to apply the large-sample techniques of the previous section. Under the assumptions mentioned

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t-distribution with $n - 1$ degrees of freedom. So, we can use the random variable T as a pivotal quantity for constructing a $100(1 - \alpha)\%$ confidence interval for μ . To do this, we start with the probability statement

$$P(-t \leq T \leq t) = 1 - \alpha,$$

where t is chosen from the t-table such that $P(T \leq -t) = P(T \geq t) = \alpha/2$. We will denote this by $t_{\alpha/2}$. So, we can write the probability statement as

$$P(-t_{\alpha/2} \leq T \leq t_{\alpha/2}) = 1 - \alpha$$

or

$$P\left(-t_{\alpha/2} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2}\right) = 1 - \alpha.$$

This probability statement can be written as

$$P\left(\bar{X} - t_{\alpha/2} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\alpha/2} \frac{S}{\sqrt{n}}\right) = 1 - \alpha,$$

which leads to a $100(1 - \alpha)\%$ confidence interval for μ ,

$$\left(\bar{X} - t_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2} \frac{S}{\sqrt{n}}\right).$$

Recall that the values of $t_{\alpha/2}$ depend on the degrees of freedom $(n - 1)$ as well as on the confidence coefficient $(1 - \alpha)$.

Under the preceding assumptions, we can also obtain $100(1 - \alpha)\%$ one-sided confidence limits for μ . Starting with the probability statement

$$P(T \leq t_{\alpha}) = 1 - \alpha,$$

leads to a lower limit interval $(\bar{X} - t_{\alpha} \frac{S}{\sqrt{n}}, \infty)$. Analogously, the probability statement

$$P(T \geq -t_{\alpha}) = 1 - \alpha,$$

leads to an upper limit confidence interval $(-\infty, \bar{X} + t_{\alpha} \frac{S}{\sqrt{n}})$.

Example 5.1 A random sample of size $n = 5$ was taken from a normal population with unknown mean and unknown variance. The observed sample values are: 2, 4, 2, 6 and 1. Develop an 80% confidence interval for population mean μ .

Solution The requirements for the t-confidence interval are satisfied. That is, we have a random sample from a normal population with mean μ and variance σ^2 . Therefore, an 80% confidence interval for μ is

$$\bar{x} \pm t_{0.2/2} \frac{s}{\sqrt{n}}.$$

Now,

$$\bar{x} = \frac{2 + 4 + 2 + 6 + 1}{5} = \frac{15}{5} = 3,$$

and

$$s^2 = \frac{(2 - 3)^2 + (4 - 3)^2 + (2 - 3)^2 + (6 - 3)^2 + (1 - 3)^2}{5 - 1} = \frac{1 + 1 + 1 + 9 + 4}{4} = \frac{16}{4} = 4.$$

From the t-table with 4 degrees of freedom, we get $t_{0.1} = 1.533$. Substituting all these into the formula, we get

$$[3 \pm 1.533(\sqrt{4/5})] = [1.629, 4.371].$$

Example 5.2 A paint manufacturer wants to determine the average drying times of a new interior wall paint. If, for 12 test areas of equal size, he obtained a mean drying time of 66.3 minutes and a standard deviation of 8.4 minutes, construct a 95% confidence interval for the true mean drying time of the new paint.

Solution Let μ denote the true mean drying time of the new paint. To apply the procedure described above, we must assume that the 12 observations constitute a random sample from a normal population (provided such assumptions are reasonable). We are given that $n = 12$, $\bar{x} = 66.3$, $s = 8.4$, and from the t-table with 11 degrees of freedom, we get $t_{0.025} = 2.201$. Substituting into the formula, we get

$$[66.3 \pm 2.201(8.4/\sqrt{12})] = [60.95, 71.64].$$

Suppose that we are interested in comparing the means of two normal populations, one with mean μ_1 and variance σ_1^2 and the other with mean μ_2 and variance σ_2^2 . If the samples are independent, confidence intervals for $\mu_1 - \mu_2$ based on a t-distributed random variable can be constructed if we assume that the two populations have a common but unknown variance, $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (unknown).

If $\bar{X}_1$ and $\bar{X}_2$ are the respective sample means obtained from independent random samples from normal populations, the large-sample confidence interval for $\mu_1 - \mu_2$ is developed by using

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

as a pivotal quantity. Because we assumed that the sampled populations are both normally distributed, Z has a standard normal distribution, and using the assumption $\sigma_1^2 = \sigma_2^2 = \sigma^2$, the quantity Z may be rewritten as

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma^2(\frac{1}{n_1} + \frac{1}{n_2})}} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Since σ^2 is not known we need to estimate it from the data. To find an estimator for the common population variance, let $X_{11}, X_{12}, \dots, X_{1n_1}$ denote the random sample of size n_1 from the first population and let $X_{21}, X_{22}, \dots, X_{2n_2}$ denote an independent random sample of size n_2 from the second population. Then

$$\bar{X}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} X_{1i} \quad \text{and} \quad \bar{X}_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} X_{2i}.$$

The usual unbiased estimator of the common variance σ^2 is obtained by pooling the sample data to obtain the *pooled estimator* S_p^2 :

$$S_p^2 = \frac{\sum_{i=1}^{n_1} (X_{1i} - \bar{X}_1)^2 + \sum_{i=1}^{n_2} (X_{2i} - \bar{X}_2)^2}{n_1 + n_2 - 2} = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

where S_i^2 is the sample variance from the i th sample, $i = 1, 2$. Notice that if $n_1 = n_2$, S_p^2 is simply the average of S_1^2 and S_2^2 . If $n_1 \neq n_2$, S_p^2 is the weighted average of S_1^2 and S_2^2 , with larger weight given to the sample variance associated with the larger sample size. Further,

$$W = \frac{(n_1 + n_2 - 2)S_p^2}{\sigma^2} = \frac{\sum_{i=1}^{n_1}(X_{1i} - \bar{X}_1)^2}{\sigma^2} + \frac{\sum_{i=1}^{n_2}(X_{2i} - \bar{X}_2)^2}{\sigma^2}$$

is the sum of two independent χ^2 -distributed random variables with $n_1 - 1$ and $n_2 - 1$ degrees of freedom, respectively. Thus, W is a chi-square random variable with $\nu = (n_1 - 1) + (n_2 - 1) = n_1 + n_2 - 2$ degrees of freedom. We now use the chi-square random variable and the *independent* standard normal random variable defined above to form a pivotal quantity

$$\begin{aligned} T &= \frac{Z}{\sqrt{\frac{W}{\nu}}} = \left[\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \right] \bigg/ \sqrt{\frac{(n_1 + n_2 - 2)S_p^2}{\sigma^2(n_1 + n_2 - 2)}} \\ &= \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}. \end{aligned}$$

We recognize this as a t-random variable with $n_1 + n_2 - 2$ degrees of freedom.

Proceeding as we did earlier in this section, we see that the confidence interval for $\mu_1 - \mu_2$ has the form

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}},$$

where $t_{\alpha/2}$ is determined from a t-distribution table with $n_1 + n_2 - 2$ degrees of freedom.

Example 5.3 To compare the durability of two paints for highway use, 10 ten-centimetre wide lines of each paint were applied across a heavily travelled road. The order was decided at random. After a period of time, each line was investigated, and received a score. The higher the score, the better the durability of the paint. The data were as follows:

Paint A	6.7	7.5	7.8	6.0	6.7	8.0	5.3	6.1	6.0	6.9
Paint B	5.0	7.9	6.9	5.7	6.2	6.4	5.5	4.6	5.8	6.0

Is there enough evidence to conclude that the two paints are of a different quality? Use $\alpha = 0.05$.

Solution For our methods to be valid, we must assume that the data constitute independent samples from normal populations with common variance σ^2 . The sample statistics are computed as

$$\bar{x}_1 = 6.7, \quad \bar{x}_2 = 6.0, \quad s_1^2 = 0.764, \quad s_2^2 = 0.884.$$

The pooled estimate of the common variance is

$$s_p^2 = \frac{(10-1)0.764 + (10-1)0.884}{10+10-2} = \frac{0.764 + 0.884}{2} = 0.824.$$

From the t-table with 18 degrees of freedom, we get $t_{0.025} = 2.101$. Substituting into the formula gives

$$(6.7 - 6.0) \pm 2.101 \sqrt{0.824 \left(\frac{2}{10} \right)} = [-0.153, 1.553].$$

This confidence interval contains both negative and positive values, so there is not enough evidence to suggest that one paint is superior to the other.

We now turn to comparing of two populations means when the samples are **not** independent, called the **paired-sample** case. In the paired-sample case, the two samples are of equal size and individual members of one sample are paired with particular members of the other sample. Suppose we have two random samples of size n ,

$$X_{11}, X_{12}, \dots, X_{1n}$$

drawn from a population with mean μ_1 and variance σ_1^2 , and

$$X_{21}, X_{22}, \dots, X_{2n}$$

drawn from a population with mean μ_2 and variance σ_2^2 . If there is some sense in which X_{1i} is paired with X_{2i} , it will usually be true that high values of X_{1i} tend to be associated with high values of X_{2i} , and low with low. For example, X_{1i} and X_{2i} might be blood pressure readings on the i th individual in a group of n , on each of two occasions. Some individual would tend to give high values on both occasions, and some low values on both occasions. In other words, there would be dependence in the two samples.

Now,

$$E(X_{1i}) = \mu_1, \quad \text{Var}(X_{1i}) = \sigma_1^2, \quad E(X_{2i}) = \mu_2, \quad \text{Var}(X_{2i}) = \sigma_2^2,$$

and

$$\text{Cov}(X_{1i}, X_{2i}) = \rho\sigma_1\sigma_2$$

where ρ is the common correlation coefficient of the variables within each pair.

Let us define

$$D_i = X_{1i} - X_{2i}, \quad i = 1, 2, \dots, n,$$

the differences between the observations within each pair. Since the pairs are assumed to be independent and identically distributed, the D_i 's are independent and identically distributed.

Now,

$$\mu_D = E(D_i) = E(X_{1i} - X_{2i}) = \mu_1 - \mu_2$$

and

$$\begin{aligned}\sigma_D^2 &= \text{Var}(D_i) = \text{Var}(X_{1i} - X_{2i}) \\ &= \text{Var}(X_{1i}) + \text{Var}(X_{2i}) - 2\text{Cov}(X_{1i}, X_{2i}) \\ &= \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2.\end{aligned}$$

From these considerations, a natural point estimator for $\mu_D = \mu_1 - \mu_2$ is the average of the differences $\bar{D} = \bar{X}_1 - \bar{X}_2$. It is seen that

$$E(\bar{D}) = \mu_D = \mu_1 - \mu_2$$

and

$$\sigma_{\bar{D}}^2 = \text{Var}(\bar{X}_1 - \bar{X}_2) = \frac{1}{n}(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2).$$

Notice that if the data had been obtained from independent samples and $n_1 = n_2 = n$, then

$$E(\bar{X}_1 - \bar{X}_2) = \mu_1 - \mu_2$$

and

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{1}{n}(\sigma_1^2 + \sigma_2^2).$$

If it is reasonable to believe that $\rho > 0$, then it is seen that $\sigma_{\bar{D}}^2 < \sigma_{\bar{X}_1 - \bar{X}_2}^2$.

Since pairing samples makes the observations within each pair dependent, we cannot use the methods previously developed. The analysis uses the n paired differences D_i , $i = 1, 2, \dots, n$. Inference regarding the differences in the means, $\mu_1 - \mu_2$, are made by making inferences regarding the mean of the differences μ_D .

Define

$$\bar{D} = \frac{1}{n} \sum_{i=1}^n D_i,$$

and

$$S_D^2 = \frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D})^2 = \frac{1}{n-1} \left[\sum_{i=1}^n D_i^2 - n\bar{D}^2 \right]$$

and employ the appropriate one-sample procedure to complete the inference. If n is small and it is reasonable to assume that the differences are approximately normally distributed, we can use inferential methods based on t-distribution. So, to construct a $100(1 - \alpha)\%$ confidence interval for μ_D , we use the pivotal quantity

$$T = \frac{\bar{D} - \mu_D}{S_D/\sqrt{n}}$$

which follows t-distribution with $n - 1$ degrees of freedom. Proceeding as before shows that the $100(1 - \alpha)\%$ confidence interval for μ_D is

$$\bar{D} \pm t_{\alpha/2} \frac{S_D}{\sqrt{n}}.$$

Example 5.4 In a small clinical trial to assess the value of new tranquillizer on psychoneurotic patients, each patient was given a week's treatment with the drug and a week's treatment with a placebo, the order being determined randomly. At the end of each week the patients had to complete a questionnaire on the basis of which he was given an anxiety score (with possible values from 0 to 30). High values corresponding to high states of anxiety. The data are as follows:

Patient	1	2	3	4	5	6	7	8	9	10
Drug	19	11	14	17	23	11	15	19	11	8
Placebo	22	18	17	19	22	12	14	11	19	7

Do the data provide sufficient evidence to conclude that the drug is useful on psychoneurotic patients? Use $\alpha = 0.05$. [Make necessary assumption]

Solution Since this is a paired-sample experiment (why?), we first reduce the data to differences and construct a confidence interval on the mean of the differences, μ_D . We get

d_i	-3	-7	-3	-2	1	-1	1	8	-8	1	-13
d_i^2	9	49	9	4	1	1	1	64	64	1	203

We get

$$\bar{d} = \frac{\sum_{i=1}^{10} d_i}{10} = \frac{-13}{10} = -1.3$$

and

$$s_d^2 = \frac{1}{10-1} \left[\sum_{i=1}^{10} d_i^2 - 10\bar{d}^2 \right] = \frac{[203 - 10(-1.3)^2]}{9} = 20.678.$$

From the t-table with 9 degrees of freedom we get $t_{0.025} = 2.262$. Substituting into the formula, we get the 95% confidence interval for μ_D

$$-1.3 \pm 2.262 \left(\sqrt{\frac{20.678}{10}} \right) = [-4.553, 1.953].$$

Since the interval encloses zero, we conclude that there is not enough evidence to show that the new drug is useful for psychoneurotic patients.

Example 5.5 Following are average weekly losses of man-hours due to accidents in 10 industrial plants before and after the installation of an elaborate safety programme:

Before	45	73	46	124	33	57	83	34	26	17
After	36	60	44	119	35	51	77	29	24	11

Do the data provide enough evidence to suggest that the safety programme is effective in reducing the average weekly losses of man-hours due to accidents? Take $\alpha = 0.05$.

Solution *Exercise*